

Project Euler

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Problem 638: Weighted Lattice Paths

Weighted Lattice Paths

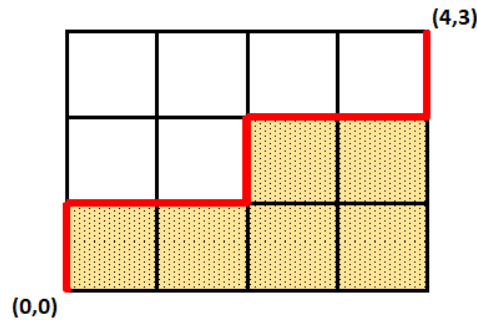
Problem 638



Let $P_{a,b}$ denote a path in a $a \times b$ lattice grid with following properties:

- The path begins at $(0, 0)$ and ends at (a, b) .
- The path consists only of unit moves upwards or to the right; that is the coordinates are increasing with every move.

Denote $A(P_{a,b})$ to be the area under the path. For the example of a $P_{4,3}$ path given below, the area equals 6.



Define $G(P_{a,b}, k) = k^{A(P_{a,b})}$. Let $C(a, b, k)$ equal the sum of $G(P_{a,b}, k)$ over all valid paths in a $a \times b$ lattice grid.

You are given that

- $C(2, 2, 1) = 6$
- $C(2, 2, 2) = 35$
- $C(10, 10, 1) = 184756$
- $C(15, 10, 3) \equiv 880419838 \pmod{1000000007}$
- $C(10000, 10000, 4) \equiv 395913804 \pmod{1000000007}$

Calculate $\sum_{k=1}^7 C(10^k + k, 10^k + k, k)$. Give your answer modulo 1000000007

Answer:

Check

Maths/Python

Let $C(a, b, k)$ be the total value of all paths from $(0, 0)$ to (a, b) .

Let A be the number of unit squares below a path.

Each path has value k^A .

For each up move, count the number of right moves before it.

This gives the number of squares below that up move.

For example, $RURU$ has area $1 + 2 = 3$.

Therefore, when $k = 2$, its value is $2^3 = 8$.

A path ending at (a, b) ends either with R or with U .

If it ends with R , it comes from $(a - 1, b)$ and adds no area.

If it ends with U , it comes from $(a, b - 1)$ and adds a to the area.

$$C(a, b, k) = C(a - 1, b, k) + k^a C(a, b - 1, k)$$

$$C(0, b, k) = 1 \quad C(a, 0, k) = 1$$

Gaussian Binomial Identity

$$G(a, b, k) = \frac{(1 - k^{b+1})(1 - k^{b+2}) \cdots (1 - k^{a+b})}{(1 - k)(1 - k^2) \cdots (1 - k^a)}$$

$$G(a - 1, b, k) = G(a, b, k) \left(\frac{1 - k^a}{1 - k^{a+b}} \right)$$

$$G(a, b - 1, k) = G(a, b, k) \left(\frac{1 - k^b}{1 - k^{a+b}} \right)$$

$$G(a - 1, b, k) + k^a G(a, b - 1, k)$$

$$= G(a, b, k) \left(\frac{1 - k^a}{1 - k^{a+b}} + k^a \frac{1 - k^b}{1 - k^{a+b}} \right)$$

$$= G(a, b, k) \left(\frac{1 - k^a + k^a - k^{a+b}}{1 - k^{a+b}} \right)$$

$$= G(a, b, k)$$

Therefore $G(a, b, k)$ has the same recurrence and starting values as $C(a, b, k)$.

Therefore $G(a, b, k) = C(a, b, k)$.

For $k = 1$, every path has value 1. Therefore,

$$C(11, 11, 1) = \frac{22!}{11! \cdot 11!} = 705432$$

The remaining values were calculated using Python modulo 1000000007.

```

1  MOD = 1_000_000_007
2
3  def weighted_paths(n, k):
4      if k == 1:
5          return 705432
6
7      numerator_product = 1
8      denominator_product = 1
9      power_n_plus_i = pow(k, n, MOD)
10     power_i = 1
11
12     for i in range(1, n + 1):
13         power_n_plus_i = (power_n_plus_i * k) % MOD
14         power_i = (power_i * k) % MOD
15
16         numerator_product = (
17             numerator_product * (power_n_plus_i - 1)
18         ) % MOD
19
20         denominator_product = (
21             denominator_product * (power_i - 1)
22         ) % MOD
23
24     inverse_denominator = pow(denominator_product, MOD - 2, MOD)
25
26     return (numerator_product * inverse_denominator) % MOD
27
28
29  answer = 0
30
31  for k in range(1, 8):
32      n = 10**k + k
33      value = weighted_paths(n, k)
34      print(f"k = {k}, n = {n}, C = {value}")
35      answer = (answer + value) % MOD
36
37  print("Answer =", answer)

```

```

k = 1, n = 11, C = 705432
k = 2, n = 102, C = 610979915
k = 3, n = 1003, C = 153114021
k = 4, n = 10004, C = 886588484
k = 5, n = 100005, C = 821855870
k = 6, n = 1000006, C = 471815489
k = 7, n = 10000007, C = 73364204
Answer = 18423394

```

Therefore the final answer is

18423394 (mod 1000000007)



Congratulations, the answer you gave to problem 638 is correct.